H

NANYANG JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9744/04

Paper 4 Practical

25 August 2017

Candidates answer on the Question Paper
Additional Materials: As listed in the Confidential Instructions

2 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do no use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

Shift
Laboratory

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do no use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **14** printed pages and **no** blank page.

[Turn over

Answer **all** the questions.

1 You are required to determine the concentration (in $mgcm^{-3}$) of ascorbic acid (Vitamin C) and glucose in a sample of orange juice.

Proceed as follows:

I. The ascorbic acid test

You have been provided with standard solutions containing **0.5, 1.0, 2.0** and **4.0** $mgcm^{-3}$ of ascorbic acid respectively. These solutions reduce the dye dichlorophenol indophenol (DCPIP) from blue to colourless:

$$\text{DCPIP (blue)} \xrightarrow{\text{ascorbic acid}} \text{reduced DCPIP (colourless)}$$

Using a syringe, place 2 cm^3 of DCPIP solution in a test tube. Place the test tube in a rack. Fill a 2 cm^3 syringe with **4.0** $\mathbf{mgcm^{-3}}$ ascorbic acid solution. Add this solution, drop by drop, to the DCPIP solution, stirring gently with a glass rod after each drop. Determine the number of drops needed to decolourise the DCPIP solution.

(a) **(i)** Note this number in the table 1 below.

Table 1

Concentration of ascorbic acid / $mgcm^{-3}$	number of drops needed to decolourise DCPIP
4.0	
2.0	
1.0	
0.5	
Orange juice	

Repeat this procedure, using fresh samples of DCPIP each time, with the other **three** solutions of ascorbic acid and, finally, with the orange juice with which you have been provided.

(ii) Add these results to the **Table 1**. [2]

(iii) Plot a graph using the data in **Table 1**. [4]

(iv) Use the data you have obtained to determine the ascorbic acid concentration in the sample of orange juice. Explain how you arrive at your answer.

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.... [2]

II. The ascorbic acid test

You have been provided with a 4% (by mass) solution of glucose, distilled water and Benedict's solution. Using only the apparatus provided, devise and carry out a procedure by which you can make the Benedict's test quantitative in order to determine the glucose concentration (in $mgcm^{-3}$) of orange juice. You should work with five different glucose solutions covering the range from 0.1% to 4% by serial dilution.

Note: In your tests you are advised to use 5 cm^3 of Benedict's solution to 0.5 cm^3 sample of all solutions.

(b) (i) Describe your method and show the results you obtained in a table.

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[5]

(ii) Estimate the glucose concentration of the orange juice in $mgcm^{-3}$.

[2]

(iii) Describe two other modifications to your method that would increase confidence in the conclusions and explain how these modifications would achieve this.

1 ....

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2 ....

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.... [4]

To a clean test-tube, add 2 cm^3 of DCPIP solution. Using a 2 cm^3 syringe as before, add the same number of drops of 4% glucose solution as you did of the 0.5 $mgcm^{-3}$ ascorbic acid (which you have recorded in **Table 1**).

(c) Note your observations.

.... [1]

Carry out Benedict's test using the 4.0 $mgcm^{-3}$ ascorbic acid solution.

(d) Note your results.

.... [1]

(e) State the significance of the procedures which you carried out in **(c)** and **(d)** for the interpretation of your results in **(a)** and **(b)**.

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[3]

[Total: 24]

2 You are required to investigate the effects of sucrose and potassium nitrate (KNO_3) solutions on the cells of the plant material supplied. Peel off several pieces of epidermis from the pigmented areas of the plant tissue, taking care to remove as little as possible of the underlying tissue. Cut the pieces of epidermis so that you have four squares of tissue each approximately 0.5 x 0.5cm. Place these in a dish of distilled water.

Take one piece of epidermis and mount it in distilled water on a microscope slide. Cover with a cover slip. Using your microscope, find an area of the tissue where pigmented cells can be seen clearly, preferably as a single layer of cells.

(a) Describe the distribution of the coloured contents within the cells.

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[1]

(b) Mount another piece of epidermis in 1 mol dm^{-3} sucrose solution. Blot off excess water with filter paper before you add the solution.

(i) After about one minute, make a large labelled drawing to show the detailed structure of one epidermal cell which is typical of the most deeply coloured cells which you can see.

[4]

(ii) Describe and explain for the appearance of the cells when placed in 1 mol dm^{-3} sucrose solution.

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.... [3]

(c) Mount another piece of epidermis in 1 mol dm^{-3} potassium nitrate solution. Immediately observe the detailed structure of a typical pigmented cell.

(i) Compare the appearance of this cell with that drawn in **(b)(i)**.

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.... [2]

(ii) Suggest a reason for the differences in the appearances of the two cells in the two solutions.

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.... [2]

(d) Several chemicals are known to affect membrane permeability.

Suggest how ethanol may have an effect on onion epidermis.

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.... [2]

(e) **Fig. 2** is a stained transverse section through part of the stem of a different plant species. You are not expected to be familiar with this specimen.

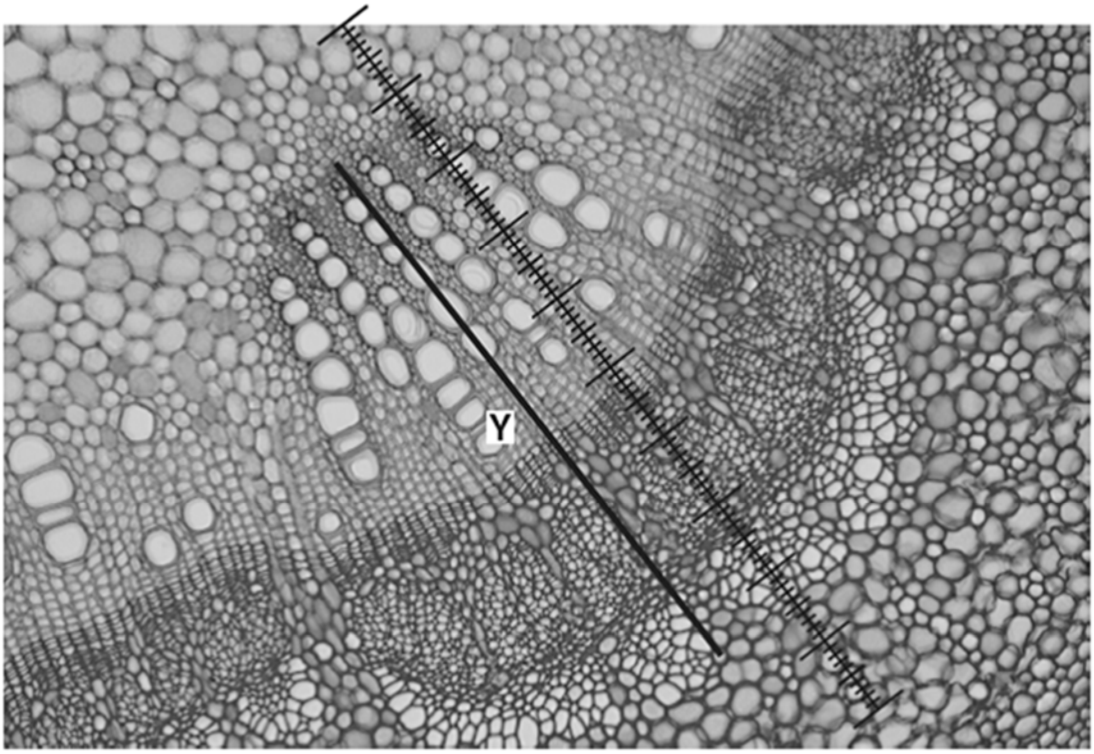

Fig. 2

A student calibrated the eyepiece graticule in a light microscope using a stage micrometer scale so that the actual length of the vascular bundle could be found.
The calibration was: one eyepiece graticule division equal to 11 µm.

Fig. 2 shows a photomicrograph taken using the same microscope with the same lenses as the student. Use the calibration of the eyepiece graticule division and **Fig. 2** to calculate the actual length of the vascular bundle, shown by line **Y**.

You may lose marks if you do not show all the steps in your working and do not use appropriate units.

[3]
[Total:17]

3 All green plants photosynthesise in the light, taking in carbon dioxide and releasing oxygen. They also respire continuously, taking in oxygen and releasing carbon dioxide. The light intensity at which photosynthesis and respiration occur at the same rate, so that there is no net gas exchange, is called the compensation point.

Compensation points can be investigated using hydrogencarbonate indicator solution. This is harmless to living organisms but changes colour over a range of concentrations of carbon dioxide due to changes in pH, as shown in Table 3.1.

Table 3.1

increasing CO_2 in indicator ←			atmospheric CO_2 level			decreasing CO_2 in indicator →		
yellow		orange		red		magenta		purple
pH 7.6	pH 7.8	pH 8.0	pH 8.2	pH 8.4	pH 8.6	pH 8.8	pH 9.0	pH 9.2

Hydrogen carbonate indicator is red in equilibrium with atmospheric air. It changes from red to magenta to deep purple as carbon dioxide concentration decreases. It changes from red to orange to yellow as carbon dioxide concentration increases

Monitoring the colour of hydrogencarbonate indicator solution in sealed vessels containing plant material can show whether carbon dioxide is being taken in or given out.

One factor that can affect the compensation point is whether a leaf is adapted for low light intensities (shade leaf) or high light intensities (sun leaf). Shade leaves would be expected to reach their compensation points at a lower light intensity than sun leaves.

Light intensity = $1/d^2$, where d represents the distance from the light source

Some plants produce both shade and sun leaves depending on where the leaves develop. For example, an aquatic plant can produce sun leaves at the top where they are in direct sunlight and produce shade leaves lower down where light intensity is reduced.

Using this information and your own knowledge, design an experiment to find the light intensity at which shade leaves and sun leaves from an aquatic plant reach their light compensation points.

Comparison of the results would then allow testing of the hypothesis that shade leaves reach their compensation points at a lower light intensity than sun leaves.

You must use:

- hydrogencarbonate indicator solution
- colourimeter and cuvette
- sun and shade leaves from an aquatic plant.

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- timer, e.g. stopwatch
- bungs
- bench lamp with 60W bulb

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to describe the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total:14]